All for one and none forall: Compiling polymorphic relations without monomorphization

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2026 August 24

All for one and none forall

Represent all concrete instances of a polymorphic relation

using only a single concrete instance

to compile away type variables

Representing relations

$$\begin{tikzcd} 0 \arrow[r, bend left] & 1 \arrow[l, bend left] & 2 \arrow[l, bend right] \arrow[r, bend left] & 3 \end{tikzcd}$$

```
(defrel (graph
  (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{array}{cc} & \text{to} \\ & \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \text{from} \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Represent graphs as relations or as adjacency matries.
semiringKanren computes matrices for arbitrary relations!

Primitive relations

`(== x y)`

$$\begin{array}{cc|c} & & y \\ & & \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ x & \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

`(=/= x y)`

$$\begin{array}{cc|c} & & y \\ & & \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ x & \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

`disj` is addition

```
(defrel (r0 (x : (Num 4)) (y : (Num 4)))
  (disj
    (== x 0)
    (== y 0)))
```

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

`conj` is multiplication

```
(defrel (r1 (x : (Num 4)) (y : (Num 4)))
  (disj
    (== x 0)
    (== y 0)))
```

$$\begin{bmatrix}1 & 1 & 1 & 1\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\end{bmatrix} * \begin{bmatrix}1 & 0 & 0 & 0\\1 & 0 & 0 & 0\\1 & 0 & 0 & 0\\1 & 0 & 0 & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\end{bmatrix}$$

fresh is summation

Vertices with incoming edges.

$$\begin{tikzcd} 0 \arrow[r, bend left] & 1 \arrow[l, bend left] & 2 \arrow[l, bend right] \arrow[r, bend left] & 3 \end{tikzcd}$$

```
(defrel (incoming-edge-vertices (y : (Num 4)))
  (fresh (x : (Num 4))
    (graph x y)))
```

$$\sum_x \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix}$$

Types

semiringKanren values inhabit finite algebraic data types.

$$
\begin{array}{lcl}
\texttt{(left 0)} & : & \texttt{(Sum (Num 2) (Num 3))} \\
\texttt{(left 1)} & : & \texttt{(Sum (Num 2) (Num 3))} \\
\texttt{(right 0)} & : & \texttt{(Sum (Num 2) (Num 3))} \\
\texttt{(right 1)} & : & \texttt{(Sum (Num 2) (Num 3))} \\
\texttt{(right 2)} & : & \texttt{(Sum (Num 2) (Num 3))}
\end{array}
\quad
\begin{array}{c}
\overbrace{\begin{matrix} & \end{matrix}}^{\text{left}}\ \overbrace{\begin{matrix} & & \end{matrix}}^{\text{right}} \\
\begin{bmatrix}1 & 0 & 0 & 0 & 0\end{bmatrix} \\
\begin{bmatrix}0 & 1 & 0 & 0 & 0\end{bmatrix} \\
\begin{bmatrix}0 & 0 & 1 & 0 & 0\end{bmatrix} \\
\begin{bmatrix}0 & 0 & 0 & 1 & 0\end{bmatrix} \\
\begin{bmatrix}0 & 0 & 0 & 0 & 1\end{bmatrix}
\end{array}
$$

(semiringKanren also supports unit and product types)

Representing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\boxed{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}} * \boxed{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Representing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\boxed{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}} * \boxed{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Representing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\boxed{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}} * \boxed{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Representing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\boxed{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}} * \boxed{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Representing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Representing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{tikzcd} 0 \arrow[r, bend left] & 1 \arrow[l, bend left] & 2 \arrow[l, bend right] \arrow[r, bend left] & 3 \end{tikzcd}$$

Introducing semiringKanren

Polymorphic semiringKanren

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\left.\begin{array}{l} x \mapsto (\texttt{right}\ \underline{2_3}) \\ y \mapsto (\texttt{left}\ \underline{2_3}) \end{array}\right\} 2_3 = 2_3 \qquad \left.\begin{array}{l} x \mapsto (\texttt{right}\ \underline{3_4}) \\ y \mapsto (\texttt{left}\ \underline{3_4}) \end{array}\right\} 3_4 = 3_4$$

x is $(\texttt{right}\ \ldots)$ $\qquad$ y is $(\texttt{left}\ \ldots)$ $\qquad$ β-typed parts equal